

A Discrete Sobolev Compactness and Weak Convergence

This appendix isolates the analytic step behind the retaliation-capacity cascade. The discrete cascade is treated as a sequence of Dirichlet Sturm–Liouville problems. The point is not to impose an Airy profile by analogy, but first to prove that near-null discrete fields have a continuum weak limit for the Sturm–Liouville operator.

Let $I = [a, b]$ and let $h = (b - a)/(n + 1)$ with grid points $x_i = a + ih$, $0 \leq i \leq n + 1$. Define the discrete Dirichlet space

$$X_h := \{u = (u_0, \dots, u_{n+1}) : u_0 = u_{n+1} = 0\}.$$

For $u, v \in X_h$, set

$$\langle u, v \rangle_{L_h^2} := h \sum_{i=1}^n u_i v_i, \quad \|u\|_{L_h^2}^2 := \langle u, u \rangle_{L_h^2},$$

and define the backward difference

$$(D_h u)_i := \frac{u_i - u_{i-1}}{h}, \quad 1 \leq i \leq n + 1.$$

The discrete H^1 norm is

$$\|u\|_{H_h^1}^2 := \|u\|_{L_h^2}^2 + h \sum_{i=1}^{n+1} |(D_h u)_i|^2.$$

The discrete Laplacian is

$$(\Delta_h u)_i := \frac{u_{i+1} - 2u_i + u_{i-1}}{h^2}, \quad 1 \leq i \leq n,$$

with the boundary values $u_0 = u_{n+1} = 0$. Given $\sigma > 0$ and a bounded grid potential $V_h = (V_{h,i})_{i=1}^n$, define

$$(H_h u)_i := -\sigma(\Delta_h u)_i + V_{h,i} u_i.$$

The dual residual is measured by

$$\|r_h\|_{H_h^{-1}} := \sup_{\varphi \in X_h, \|\varphi\|_{H_h^1} = 1} \left| h \sum_{i=1}^n r_{h,i} \varphi_i \right|.$$

Let $I_h u$ be the continuous piecewise-linear interpolant of u with nodal values u_i at x_i . The elementary interpolation estimates used below are

$$c_0 \|u\|_{H_h^1} \leq \|I_h u\|_{H_0^1(I)} \leq C_0 \|u\|_{H_h^1}, \quad (\text{A.1})$$

with constants independent of h . The Dirichlet boundary also gives a uniform discrete Poincare inequality,

$$\|u\|_{L_h^2} \leq C_P \left(h \sum_{i=1}^{n+1} |(D_h u)_i|^2 \right)^{1/2}. \quad (\text{A.2})$$

If one works with Neumann data or conditional measures, the analogous statement is a quotient estimate modulo constants; here the zero boundary data remove that quotient by fixing the centered representative.

Theorem 1 (Discrete Cascade-to-Airy Convergence). *Let $\sigma > 0$. Suppose $V \in C([a, b])$ and the grid potentials satisfy*

$$\max_{1 \leq i \leq n} |V_{h,i} - V(x_i)| \rightarrow 0, \quad \sup_{h,i} |V_{h,i}| < \infty.$$

Let $\psi_h \in X_h$ be a sequence of discrete retaliation-capacity fields such that

$$\|\psi_h\|_{L_h^2} = 1, \quad \|H_h \psi_h\|_{H_h^{-1}} \rightarrow 0.$$

Then, after passing to a subsequence, $I_h \psi_h$ converges weakly in $H_0^1(I)$ and strongly in $L^2(I)$ to a non-zero field $\Psi \in H_0^1(I)$ with $\|\Psi\|_{L^2(I)} = 1$. The limit solves the weak Sturm-Liouville equation

$$\sigma \int_a^b \Psi'(x) \eta'(x) dx + \int_a^b V(x) \Psi(x) \eta(x) dx = 0 \quad \text{for every } \eta \in H_0^1(I). \quad (\text{A.3})$$

Equivalently, $-\sigma \Psi'' + V \Psi = 0$ in $H^{-1}(I)$. Near any simple turning point of V , this weak limit has the Airy normal form described in Appendix B.

Proof. Write the discrete bilinear form associated with H_h as

$$a_h(u, \varphi) := \sigma h \sum_{i=1}^{n+1} (D_h u)_i (D_h \varphi)_i + h \sum_{i=1}^n V_{h,i} u_i \varphi_i.$$

Discrete summation by parts gives

$$a_h(u, \varphi) = h \sum_{i=1}^n (H_h u)_i \varphi_i, \quad u, \varphi \in X_h. \quad (\text{A.4})$$

The potential may change sign, so the unshifted form is not assumed coercive. Choose $\gamma > 1 + \sup_{h,i} |V_{h,i}|$ and define the shifted form

$$a_h^\gamma(u, u) := a_h(u, u) + \gamma \|u\|_{L_h^2}^2.$$

Then

$$a_h^\gamma(u, u) \geq \sigma h \sum_{i=1}^{n+1} |(D_h u)_i|^2 + \|u\|_{L_h^2}^2, \quad (\text{A.5})$$

after decreasing the right-hand constant if necessary. This is the Garding shift: the positive shift supplies coercivity without claiming that the sign-changing operator H_h is coercive on its own.

Using (A.4), the residual bound, and $\|\psi_h\|_{L_h^2} = 1$,

$$a_h^\gamma(\psi_h, \psi_h) = h \sum_{i=1}^n (H_h \psi_h)_i \psi_{h,i} + \gamma \leq \|H_h \psi_h\|_{H_h^{-1}} \|\psi_h\|_{H_h^1} + \gamma.$$

Together with (A.5), this quadratic inequality gives a uniform H_h^1 bound on ψ_h . By the interpolation equivalence (A.1), the piecewise-linear fields $I_h \psi_h$ are uniformly bounded in $H_0^1(I)$. Rellich compactness therefore gives a subsequence, not relabeled, and $\Psi \in H_0^1(I)$ such that

$$I_h \psi_h \rightharpoonup \Psi \quad \text{in } H_0^1(I), \quad I_h \psi_h \rightarrow \Psi \quad \text{in } L^2(I).$$

The normalization transfers to the limit because the discrete L_h^2 norm and the L^2 norm of the piecewise-linear interpolant differ by $o(1)$ on uniformly H_h^1 -bounded sequences. Hence $\|\Psi\|_{L^2(I)} = 1$.

It remains to pass to the weak equation. Fix $\eta \in C_c^\infty(a, b)$ and let $\eta_h \in X_h$ be its nodal restriction, $\eta_{h,i} = \eta(x_i)$. The residual assumption gives

$$a_h(\psi_h, \eta_h) = h \sum_{i=1}^n (H_h \psi_h)_i \eta(x_i) \rightarrow 0. \quad (\text{A.6})$$

For the gradient part,

$$h \sum_{i=1}^{n+1} (D_h \psi_h)_i (D_h \eta_h)_i = \int_a^b (I_h \psi_h)'(x) (I_h \eta_h)'(x) dx \rightarrow \int_a^b \Psi'(x) \eta'(x) dx,$$

using weak convergence in H^1 and the strong convergence $I_h \eta_h \rightarrow \eta$ in $H_0^1(I)$. For the potential term, uniform convergence of V_h to V , strong L^2 convergence of $I_h \psi_h$, and the consistency of nodal quadrature for the smooth test function give

$$h \sum_{i=1}^n V_{h,i} \psi_{h,i} \eta(x_i) \rightarrow \int_a^b V(x) \Psi(x) \eta(x) dx.$$

Taking limits in (A.6) proves (A.3) for smooth compactly supported tests. Density of $C_c^\infty(a, b)$ in $H_0^1(I)$ extends the identity to every $\eta \in H_0^1(I)$. $\square$

B Airy Normal Form near a Simple Turning Point

The Airy profile appears only after the continuum weak equation has been obtained. Assume that $V \in C^2$ in a neighborhood of x_c and that

$$V(x_c) = 0, \quad V'(x_c) = -c < 0. \quad (\text{B.1})$$

By one-dimensional elliptic regularity applied to $-\sigma\Psi'' + V\Psi = 0$, the weak limit from Theorem 1 is C^2 locally near x_c . Taylor expansion gives

$$V(x) = -c(x - x_c) + O((x - x_c)^2).$$

Set

$$z := \left(\frac{c}{\sigma}\right)^{1/3} (x - x_c), \quad Y(z) := \Psi\left(x_c + \left(\frac{\sigma}{c}\right)^{1/3} z\right).$$

Then the local equation becomes

$$Y''(z) + zY(z) = O(z^2)Y(z). \quad (\text{B.2})$$

Keeping the first-order turning-point term gives the Airy normal form

$$Y''(z) + zY(z) = 0. \quad (\text{B.3})$$

Its local solution space is two-dimensional:

$$Y(z) = A \operatorname{Ai}(-z) + B \operatorname{Bi}(-z). \quad (\text{B.4})$$

Equation (B.4) is a local normal form, not a branch selection rule. The pure $\operatorname{Ai}(-z)$ branch is justified only after an additional matching condition is imposed, for example finite-energy matching from the protected side or a boundary condition that eliminates the growing $\operatorname{Bi}(-z)$ component. Without that matching information, the discrete cascade theorem identifies the continuum Sturm–Liouville limit and the local Airy basis, but it does not select $B = 0$.